ABSTRACT

The Crop Advisory and Disease Detection System is a web-based platform designed to support farmers and agricultural students in diagnosing crop diseases and accessing relevant crop information based on soil and environmental factors. It offers a centralized, user-friendly interface that consolidates various agricultural services for ease of use. The system enables users to upload images of crop leaves for disease identification. Its backend analyses these images by extracting visual features and comparing them with a database of known disease patterns to identify the closest match. Following diagnosis, the system provides recommended preventive measures and management suggestions to assist users in crop care. Beyond disease detection, the platform features a crop advisory module that supplies detailed information about different crops, including suitable soil types, basic descriptions, and common diseases. Users can retrieve this data through direct searches or system-generated recommendations. The system also includes a ticket-based support module, enabling users to submit queries and receive expert guidance. All data covering user profiles, disease records, crop details, geographic information, weather data, and support requests is stored in a centralized database, ensuring efficient access, data consistency, and system reliability.

INTRODUCTION

The Crop Advisory and Disease Detection System is a web-based digital solution designed to assist farmers and agriculture students in identifying crop diseases and accessing relevant crop-related information efficiently. This system plays a crucial role in improving agricultural practices by providing structured and easily accessible information related to crop health, soil suitability, and environmental conditions. Traditionally, accessing such information required manual observation or consulting experts, which could be time-consuming and less organized. With the implementation of this system, these challenges are reduced, ensuring a more efficient and systematic approach to agricultural management. The primary objective of the Crop Advisory and Disease Detection System is to provide an effective platform where users can identify crop diseases and access useful crop information. The system enables users to upload images of crop leaves for analysis. It processes the input by extracting visual features and comparing them with stored datasets to identify the closest matching In addition, the system includes a crop advisory module that allows users to search for crops and view details such as suitable soil type, basic description, and commonly associated diseases. The system also integrates a location-based weather feature, where users can select a location using a map interface and view real-time weather information. This helps users understand environmental conditions that may influence crop growth and agricultural planning. Furthermore, the system maintains a centralized database that stores user information, disease records, crop details, weather-related data, and support queries. This enables efficient data management and quick retrieval of information. The system also includes a ticket-based support module, allowing users to raise queries and receive guidance when required. disease. Based on the result, the system provides basic preventive measures and suggestions to support crop health management.

OBJECTIVES

- A web-based application that allows users to upload crop images and identify possible diseases.
- A crop advisory module that provides information such as suitable soil type, crop description, and associated diseases.
- A location-based weather feature that displays real-time weather information using a map interface.
- A centralized database system to store user data, disease records, crop details, and support queries.
- A ticket-based support system for users to raise queries and receive guidance.
- A responsive design that ensures the system can be accessed on desktops, tablets, and mobile devices.
- A user-friendly interface that allows easy navigation and interaction with the system

EXISTING SYSTEM

Difficulty in Disease Identification: Users may find it hard to accurately identify crop diseases based on visual observation alone. Lack of Organized Crop Information: Information about crops such as suitable soil type, description, and associated diseases is not easily accessible in a structured form. Limited Access to Environmental Data: Users may not have quick access to location-based weather information that affects crop growth. Manual Dependency: Users often rely on traditional methods or external help to understand crop conditions, which can be time consuming. Inefficient Data Management: Storing and accessing crop, disease, and user-related data without a proper system can be difficult and unorganized. Lack of Support System: Users may not have a structured way to raise queries and receive guidance regarding crop-related issues.

PROPOSED SYSTEM

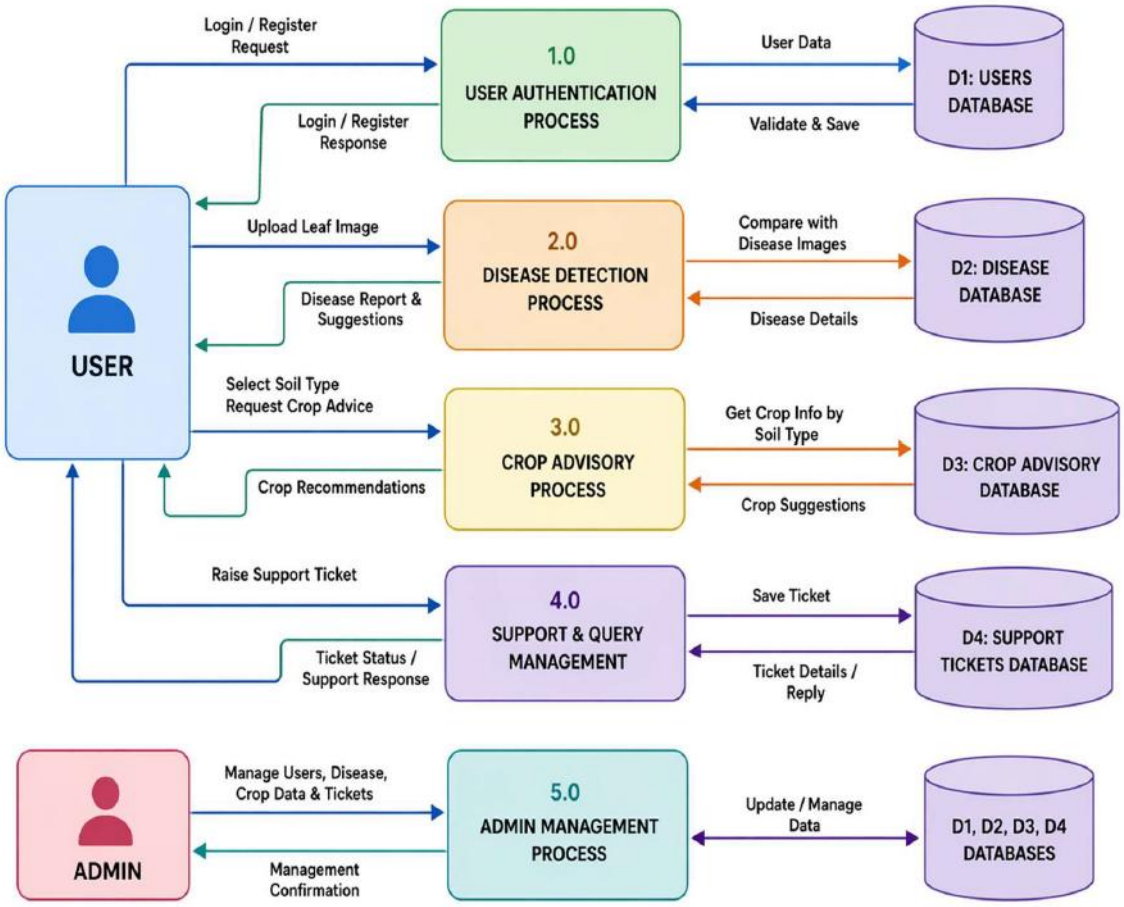
The Crop Advisory and Disease Detection System is a web-based application designed to simplify and streamline access to agricultural information for farmers and agriculture students. The system allows users to upload images of crop leaves to identify possible diseases and access relevant crop related information efficiently. It also enables users to search for crops, view details such as suitable soil type and description, and understand commonly associated diseases.

The system includes a location-based weather feature, where users can select a location using a map interface and view real-time weather information. This helps users understand environmental conditions that may affect crop growth. In addition, a ticket-based support module allows users to raise queries and receive guidance when required.

The system also maintains a centralized database to store user information, disease records, crop details, weather data, and support queries. With a user-friendly interface and structured data management, the system reduces effort, improves accessibility, and supports better decision-making in agricultural practices.

DATAFLOW DIAGRAM

The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.



MODULES

Admin Module

User Management: This module allows to handle user registrations and manage user accounts within the system.

Data Management: This feature enables to manage crop data and disease-related information used for detection and advisory.

System Configuration: This module enables administrators to customize the system settings according to their needs.

Support Management: Effective communication is maintained by allowing administrators to handle user queries through the ticket-based support system.

User Module

Profile Management: This feature allows users to manage their personal details and other relevant information.

Image Upload: Users can upload crop leaf images directly through the system for disease identification.

Crop Advisory Access: Users can search and view crop-related information including soil suitability, descriptions, and associated diseases.

Weather Access: Users can access location-based weather information through the map interface.

User Support: This feature allows users to raise queries or request assistance through the system and view responses.

System Processing Module

- The system processes uploaded images by extracting visual features such as colour ratios and brightness.
- The system compares extracted features with stored disease data to identify the closest matching disease.
- The system generates outputs including disease results and basic preventive suggestions.
- The system maintains records of user activities, disease results, crop data, and support queries in a centralized database.

SPECFICATIONS

SOFTWARE REQUIREMENTS

Processor : i3 or more

Hard Disk : 80 GB

RAM : 8GB or more

SOFTWARE REQUIREMENT

Operating System : Windows 10/11

User Interface : Thymeleaf, HTML5, CSS

Client-side Scripting : JavaScript

Programming Language : JAVA

Framework : Spring Boot

Database : MYSQL

CONCLUSION

The Crop Advisory and Disease Detection System is designed to improve agricultural decision-making by providing intelligent, technology-driven solutions for farmers and users. The system enhances the process of identifying crop diseases, accessing crop-related information, and understanding environmental conditions affecting crop growth. By integrating modern technologies, the system overcomes the limitations of traditional farming practices, offering efficient solutions such as image-based disease detection, crop recommendations, weather updates, and a support system for expert guidance.

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